



Proactive Inpatient Bed Requests for Emergency Department Admissions

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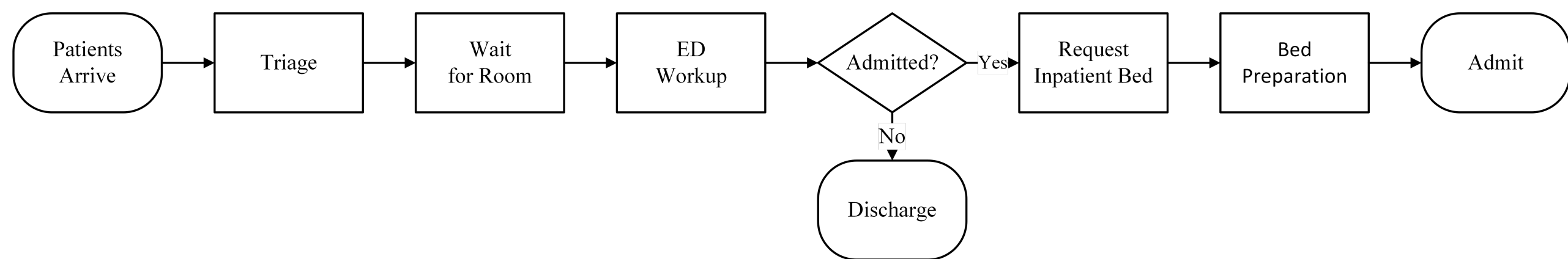
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1. Motivation

- A **boarded patient** remains in the ED after an admission decision, waiting for an inpatient bed.
- Boarding drives ED crowding, adverse outcomes, and **costs up to 2× more per patient** than inpatient care.
- In our partner ED, admitted patients wait an average of **over 4 hours** after their admission decision under current practice.

Root cause: The Transfer Preparation Process — identifying a bed, notifying staff, and preparing for transfer — begins only *after* the admission decision.



- Individual** early requests (Chen et al. 2022): bed requested per patient at triage if “highly likely” to be admitted. High single-patient uncertainty.
- Our approach:** hourly aggregate requests at the ED level. Lower variability, easier for hospital management, better aligned with operational practice.

2. Problem Formulation

State: $\mathbf{s} = (n_1, \dots, n_M, k, r)$

- n_m : # class- m patients in ED workup
- $k > 0$: boarding patients; $k < 0$: idle prepared beds
- r : beds currently under preparation

Action $a \in \{0, 1, \dots, A\}$: beds to request this hour.

Per-period cost: $C(\mathbf{s}, a) = \underbrace{a^2}_{\text{variability penalty}} + \underbrace{c_b \max(0, k)}_{\text{boarding cost}} + \underbrace{c_e \max(0, -k)}_{\text{bed idling cost}}$

Minimize expected total discounted cost over an infinite horizon. Tuning (c_b, c_e) lets management balance boarding reduction against idle bed preparation.

Performance metrics:

- LoS:** Length of stay, i.e. avg. time from ED arrival to departure (all patients)
- Boarding time:** avg. time admitted patients wait in ED after admission decision
- Bed idling time:** avg. time a prepared bed sits empty before being occupied
- CoV:** coefficient of variation of hourly bed demand (smoothness of the request process)

Benchmark policies:

- Current Practice (CP):** Request a bed only after the admission decision. Zero idling, maximum boarding.
- Greedy Policy (GP):** Request a number of beds equal to the difference between the expected number of current ED patients who will be admitted and the number of beds already requested and outstanding. No tuning parameter.
- Extreme Policy (EP):** Every patient treated as a certain admit. Theoretical lower bound on LoS.

3. Benchmark Policy Performance

Simulation results under Standard Operating Conditions:

Policy	LoS (min)	Boarding (min)	Idle (min)
CP	406.6	248.3	0.0
GP	349.0	95.8	34.6
EP	342.2	76.7	356.7

- CP** eliminates bed idling entirely but incurs maximum boarding time.
- GP** reduces LoS by ≈ 58 min (14%) and boarding by ≈ 152 min (61%) relative to CP with no tuning parameter; a strong simple baseline.
- EP** sets the theoretical lower bound on achievable LoS, but at the cost of operationally infeasible bed idling (≈ 357 min).

4. Proposed Policies

MHP: MDP-Based Heuristic Policy

- The full MDP is intractable; we analytically solve a simpler auxiliary MDP and substitute its cost-to-go into the original.
- Collapse state to a single *imbalance* scalar:

$$\phi_t = \sum_m (\alpha_m n_{m,t} + k_t - r_t) \begin{cases} > 0, & \text{if demand outpaces supply;} \\ < 0, & \text{if supply outpaces demand.} \end{cases}$$

- For the auxiliary MDP, the optimal policy is **linear** in ϕ_t and the optimal value function is **quadratic** in ϕ_t .
- Transition probabilities are estimated via Monte Carlo sampling of future states, allowing non-stationary arrivals and time-varying service rates to be incorporated without analytical intractability.
- Substituting back yields a tractable heuristic; parameters $(\tilde{\alpha}, \tilde{\beta})$ tuned on the Python simulator.

NHP: Newsvendor Heuristic Policy

- Treat each hour as a one-shot inventory problem.
- Demand** over next ψ hours = expected admissions minus beds already becoming available. Request quantity solves:

$$\min_{a \geq 0} \{a^2 + \tilde{c}_b \mathbb{E}[\text{unmet demand}] + \tilde{c}_e \mathbb{E}[\text{excess beds}]\}$$

- Mean and variance of demand computed in closed form from current state.
- Demand approximated as symmetric triangular $\Rightarrow$ **closed-form** a^* , updated every hour.
- Most transparent heuristic; no simulator needed at decision time.
- Parameters: $\tilde{c}_b, \tilde{c}_e, \psi = 8$ hours.

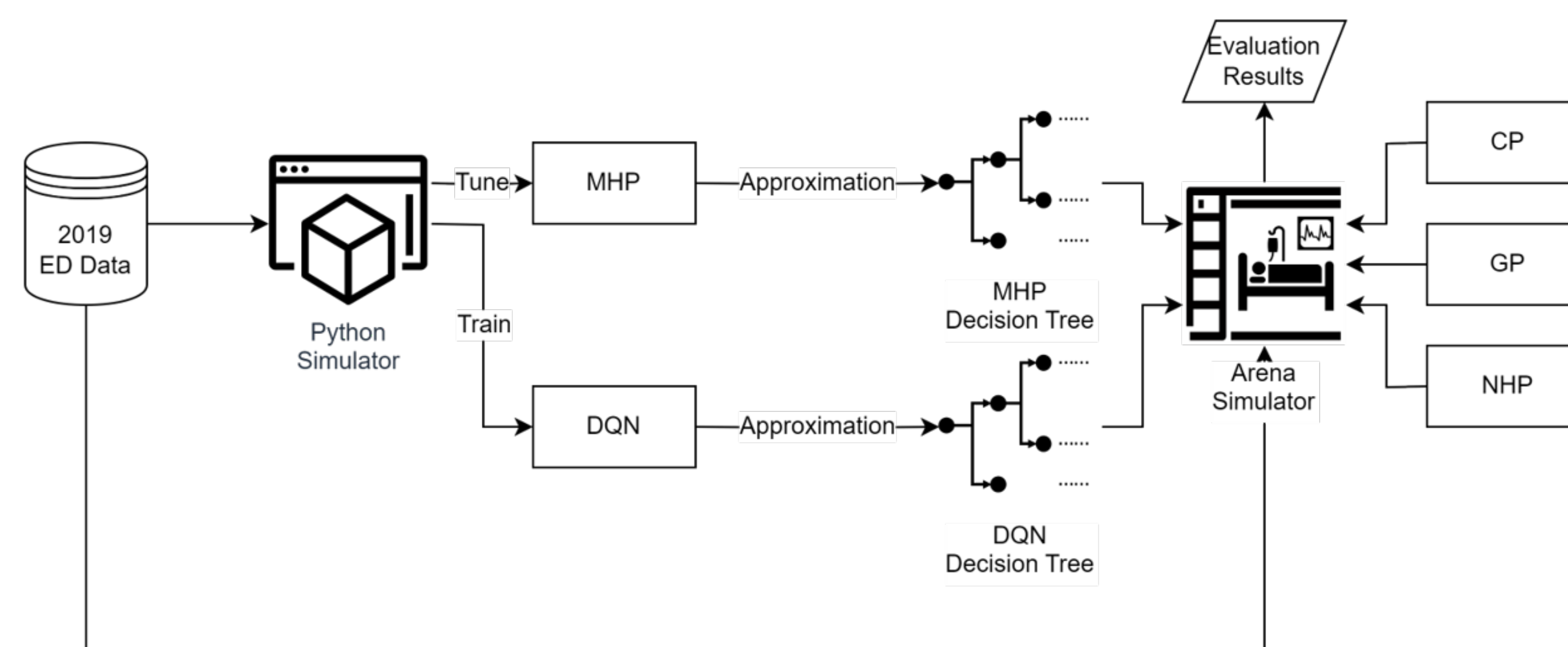
DQN: Deep Q-Learning Policy

- Agent interacts with the Python simulator to learn $Q(\mathbf{s}, a)$, the expected total discounted cost of action a followed by optimal behavior:

$$a^*(\mathbf{s}) = \arg \min_a Q(\mathbf{s}, a)$$

- A neural network approximates Q ; weights updated to minimize squared prediction error.
- Experience replay:** training on randomly sampled past transitions reduces update correlation and improves stability.
- Double-DQN** architecture with slowly-updated target network reduces Q-value overestimation.
- Policy approximated by a decision tree for interpretable deployment in Arena.

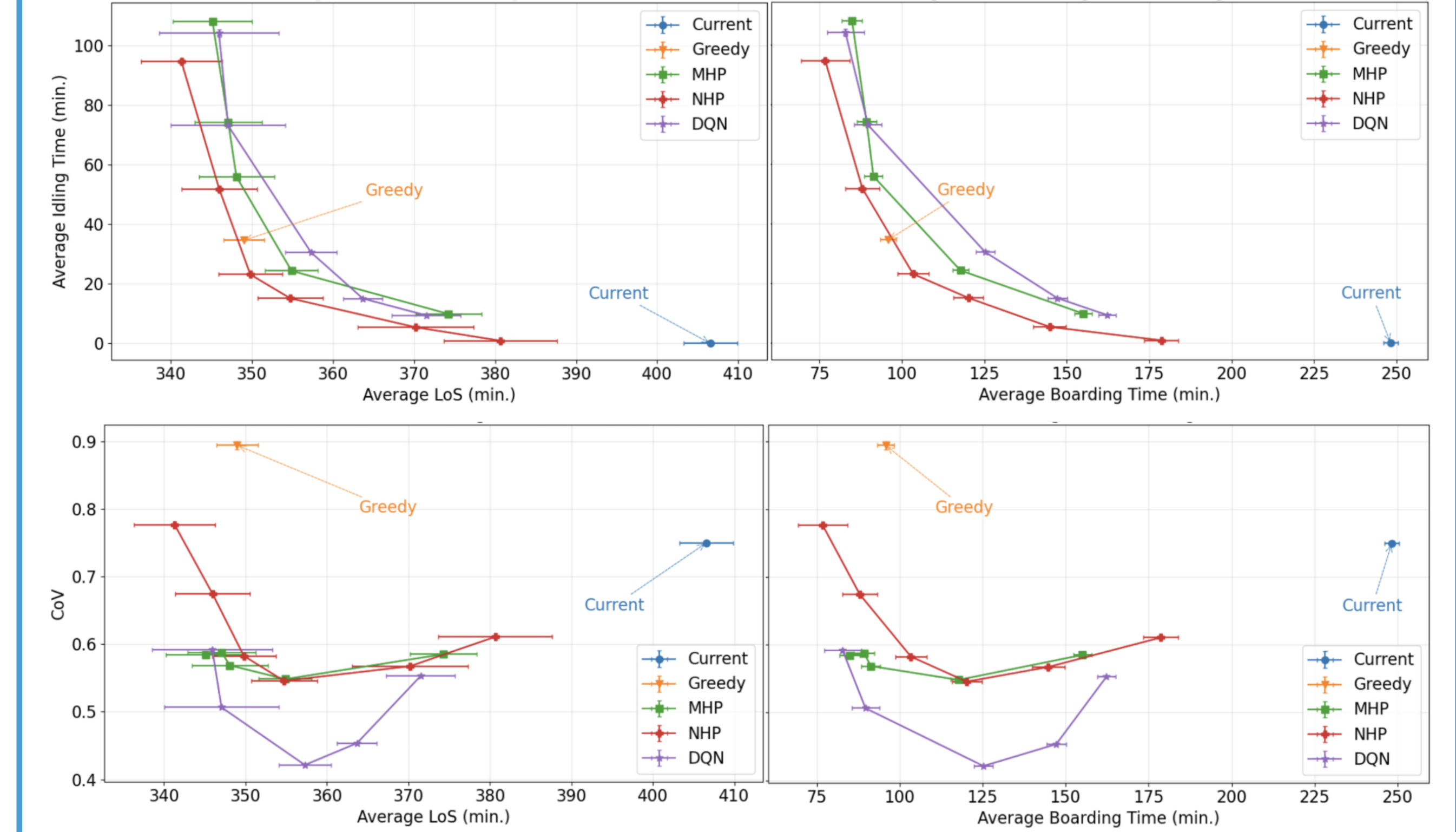
5. Simulation Framework



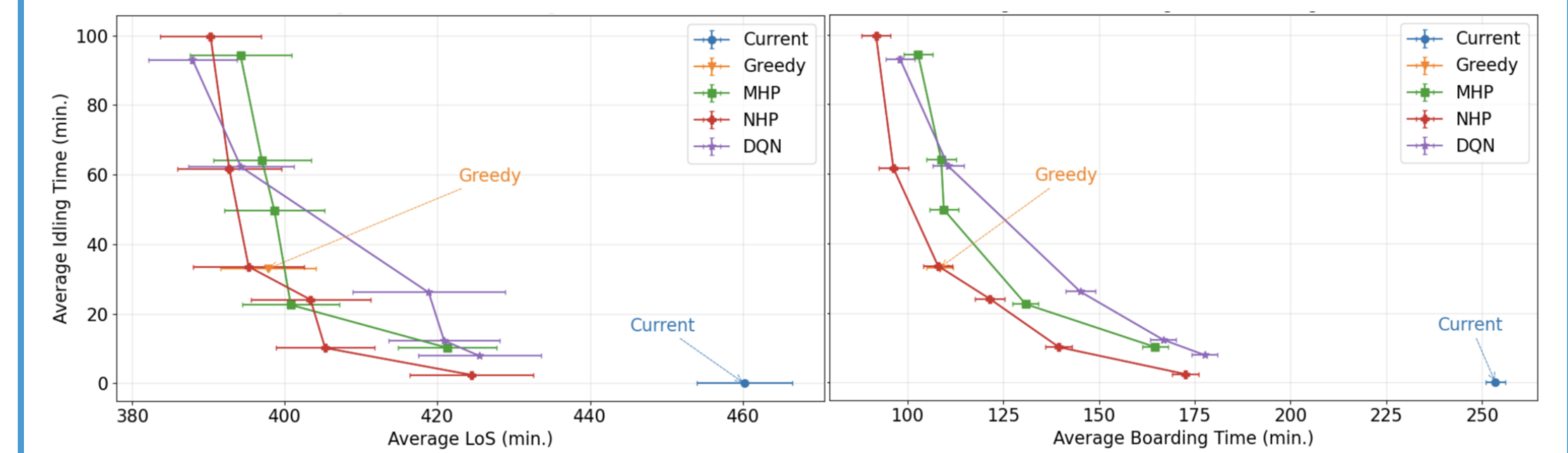
- A **Python simulator** (low granularity, fast) is used to train and tune the heuristics.
- An **Arena simulator** (high granularity, uses 2019 ED data) serves as the evaluation testbed.
- MHP and DQN policies are approximated by pruned **decision trees** (≈ 300 nodes) for interpretable deployment in Arena.
- State-action pairs (1.2 million per heuristic) are collected under each tuned policy and used to train pruned decision trees via cost-complexity pruning.

6. Results

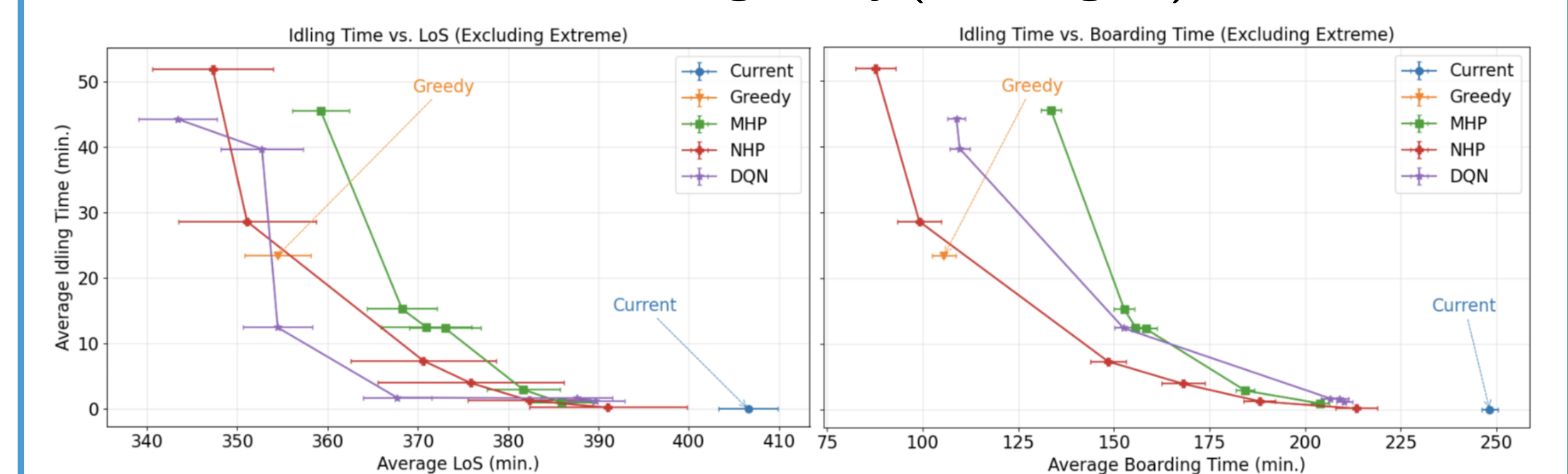
Standard Operating Scenario:



Pandemic Response Scenario:



Partial Interchangeability (70% eligible):



7. Key Takeaways & Future Work

- Aggregate early bed requests cut boarding by **30–70%** and ED LoS by **6–16%** (compared to Current Policy) with modest bed idling.
- NHP** is most interpretable and best on the LoS/idling frontier.
- MHP** achieves the largest LoS reductions at aggressive parameter settings, nearly matching the theoretical lower bound.
- DQN** produces the smoothest request patterns ensuring a stable downstream.
- Results are robust to partial bed interchangeability and pandemic-level arrival surges.
- Framework is tunable to management priorities via (c_b, c_e) .

Future work:

- Controlled pilot study at a partner hospital
- Extensions to multiple bed types (ICU, step-down, surgical)
- Online learning to adapt to seasonal and structural shifts



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